2.2.4 Multimeter

In both robotics and electronics, you need to measure three things very frequently:

A ceramic capacitor and Symbol of a non polar capacitor

Continuity between two points

To measure continuity, you should set the multimeter knob to continuity. Then, insert or place the probes of the multimeter in between the points to be checked for continuity. If you hear a beep, that means there is a connection between those two points.

Resistance between two points

Turn the knob to point towards the resistance measurement section and select upper value of the resistance randomly. Connect the probes across the points you want to measure the resistance. The resistance is displayed in ohms. If you see a 1, it means that the upper value (or limit) you selected is not appropriate, and the value of the resistance being measured is greater than it. In that case, you need to select a higher upper limit.

For example, if you want to measure a resistance and you set the knob to 20 kilo ohm. If the value of resistance is smaller than 20 kilo ohm, the multimeter screen will give you the value. But if the value of the resistance to be measured is greater than 20 kilo ohm, the multimeter will show 1, and in that case you need to select the 200 kilo ohm limit.

You need a multimeter through the building and troubleshooting stages

Voltage difference between two points

To check DC voltage difference between two points, set the knob to point in the direction of voltage measurement and hold the probes across the points you want to measure voltage. The upper limit is selected like in the case of resistance.

If the screen shows a voltage value along with a negative sign, it means the polarity is reversed. Interchange the red and black cables and the polarity will be positive.

And all the aforementioned quantities can easily be measured using a multimeter. The cost of a multimeter increases with the number